

Lecture 10: Thermal Properties of Matter – Supplementary Notes

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These notes support Lecture 10. They contain background that was moved out of the contact time. Sections marked (extension) are not assessed.

1 Pressure units and gauge pressure

1 atm = 101.325 kPa = 760 mmHg $\approx$ 14.7 psi and 1 bar = 10^5 Pa. Tyre gauges, cylinder gauges and blood-pressure monitors read the *gauge* pressure, $p - p_{\text{atm}}$. The ideal-gas equation and the kinetic theory refer to the absolute pressure, the force per area exerted by the gas itself.

2 The ideal gas as a model, and its limits (extension)

The ideal gas has point molecules and no forces between them. Real molecules have a finite size and attract one another at a distance. The van der Waals equation corrects for both:

$$\left(p + \frac{an^2}{V^2}\right)(V - nb) = nRT,$$

where b is the volume excluded by the molecules (per mole) and a measures the attraction. For nitrogen, $a \approx 0.137 \text{ Pa m}^6/\text{mol}^2$ and $b \approx 3.9 \times 10^{-5} \text{ m}^3/\text{mol}$. For 1 mol in 1.0 L at 300 K the ideal-gas pressure is $2.49 \times 10^6 \text{ Pa}$ and the van der Waals pressure is $2.46 \times 10^6 \text{ Pa}$, about 1.5% lower; in 10 L the difference is 0.16%. The ideal-gas equation is therefore a very good model at low density and a poor one near condensation, where the isotherms of Contact 2 develop a flat liquid-vapour section.

3 The pressure derivation in more detail

The cube is a convenience: the result $p = \frac{1}{3}(N/V)m\langle v^2 \rangle_{\text{av}}$ holds for a container of any shape. Collisions between molecules do not change the result: they exchange momentum and energy between molecules, but leave the average of v_x^2 unchanged because the number of molecules at each speed stays the same in equilibrium. Using only the time $2L/v_x$ between wall collisions neglects those collisions, and the correct result is the same. The factor $\frac{1}{3}$ appears because the motion is random: $\langle v_x^2 \rangle_{\text{av}} = \langle v_y^2 \rangle_{\text{av}} = \langle v_z^2 \rangle_{\text{av}}$ and their sum is $\langle v^2 \rangle_{\text{av}}$.

4 The Maxwell speed distribution (extension)

The fraction of molecules with speeds between v and $v + dv$ is $f(v) dv$ with

$$f(v) = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} v^2 e^{-mv^2/2kT}.$$

The three characteristic speeds are

$$v_{\text{mp}} = \sqrt{\frac{2kT}{m}}, \quad v_{\text{av}} = \sqrt{\frac{8kT}{\pi m}}, \quad v_{\text{rms}} = \sqrt{\frac{3kT}{m}},$$

in the ratio 1 : 1.128 : 1.225. For N_2 at 300 K: $v_{mp} = 422$, $v_{av} = 476$ and $v_{rms} = 517$ m/s. The factor $e^{-mv^2/2kT}$ falls off rapidly at high speed but never reaches zero, which is the tail referred to in the Earth-atmosphere extension.

5 Mean free path (extension)

A molecule of radius r travels, on average, a distance $\lambda = \frac{kT}{4\pi\sqrt{2}r^2p}$ between collisions. For nitrogen at 20°C and 1 atm, with $r \approx 2.0 \times 10^{-10}$ m, $\lambda \approx 5.6 \times 10^{-8}$ m, about 150 molecular diameters; the time between collisions is of order 10^{-10} s. At low pressure λ is large, which is why a vacuum chamber behaves almost as the kinetic model assumes.

6 Why latent heat exists

In a solid the molecules are held in a regular arrangement by attractive forces; in a liquid they are still close but free to move past each other; in a gas they are far apart. Melting needs energy to loosen the arrangement, and vaporization needs much more to separate the molecules completely, so $L_v > L_f$. During the change the *average kinetic energy per molecule*, and hence the temperature, is unchanged ($K = \frac{3}{2}kT$); the energy goes into the potential energy of the molecular arrangement. This distinction between kinetic and potential energy of the molecules leads to the internal energy of Lecture 11.

7 Critical point, triple point and the water phase diagram

For water the triple point is 273.16 K and 610 Pa (0.006 atm), and the critical point is 647.4 K and 2.21×10^7 Pa (218 atm). For carbon dioxide the triple point is 216.55 K and 5.17×10^5 Pa (5.1 atm), above atmospheric pressure, so solid CO_2 sublimates at 1 atm. Water is unusual: ice is less dense than liquid water, so the fusion curve slopes slightly backward (higher pressure lowers the melting temperature). Replotting the isotherms of a real gas as a surface over the p, V, T axes gives the pVT surface, whose projections are the pV and pT diagrams.

8 Equipartition (preview of Lecture 11)

The result $K = \frac{3}{2}kT$ is the energy of three independent translational motions: $\frac{1}{2}kT$ for each. The equipartition theorem states that each independent quadratic term in the energy of a molecule has an average of $\frac{1}{2}kT$ in equilibrium. A monatomic gas has only the three translational terms, so its energy per mole is $\frac{3}{2}RT$; a diatomic molecule at room temperature also has two rotational terms, giving $\frac{5}{2}RT$. These numbers set the molar heat capacities of gases, which we meet with the first law.